

LECTURE 2: INTEGERS $\mathbb{Z}$ AND RATIONAL NUMBERS $\mathbb{Q}$ **1. The set of Integers $\mathbb{Z}$.**

We assume that the set of natural numbers $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$ is given. We also assume that we know all usual properties and structures. Specifically, $\mathbb{N}$ comes with addition and multiplication which satisfy usual properties like $a + b = b + a$ for every $a \in \mathbb{N}$ and every $b \in \mathbb{N}$. The operation of subtraction cannot always be performed within natural numbers. For example, the difference $1 - 2$ is not a natural number. Let's consider this operation in some detail.

Subtraction is defined as the inverse operation to addition. By definition, $a - b$ is a number x such that $x + b = a$. In other words, $a - b$ is a solution to the equation

$$x + b = a \quad (0.1)$$

The absence of a natural number $1 - 2$ translates to the absence of solutions for $x + 2 = 1$. We construct integers to supply equations (0.1) with solutions.

Underlying Idea: An integer should be a solution to (0.1). Every such equation is determined by an ordered pair of natural numbers (a, b) . However, different equations may have the same solution. For instance, the equations $x + 1 = 2$ and $x + 3 = 4$ have the same solution $x = 1$. Thus, an integer should be a set of these equations, or equivalently, a set of ordered pairs (a, b) . This idea motivates our construction.

Definition 1. Consider the set $\mathbb{N} \times \mathbb{N}$ of ordered pairs of natural numbers. Consider the relation $\sim$ on this set defined by

$$(m, n) \sim (p, q) \iff m + q = n + p \quad (0.2)$$

Proposition 2. The relation $\sim$ is an equivalence relation.

Proof: Exercise. Check the relation satisfies:

- 1) The relation is reflexive, that is $(p, q) \sim (p, q)$.
- 2) The relation is symmetric, that is, if $(p, q) \sim (n, m)$ then $(n, m) \sim (p, q)$.
- 3) The relation is transitive, that is, if $(p, q) \sim (n, m)$ and $(n, m) \sim (r, s)$, then $(p, q) \sim (r, s)$.

□

We define the set of integers as

$$\boxed{\mathbb{Z} = \mathbb{N} \times \mathbb{N} / \sim} \quad (0.3)$$

An integer is an equivalence class $[(m, n)]$, conceptually representing the difference $m - n$. We identify an integer $n \in \mathbb{N}$ with the equivalence class $[(n, 0)]$, as n is the solution of $x + 0 = n$. This identification allows us to consider $\mathbb{N}$ as a subset of $\mathbb{Z}$.

1.1 Addition and Multiplication:

We define usual operations on integers as follows:

addition +

$$[(m, n)] + [(p, q)] = [(m + p, n + q)] \quad (0.4)$$

multiplication ·

$$[(m, n)] \cdot [(p, q)] = [(mp + nq, np + mq)] \quad (0.5)$$

Well Definedness: The definition of addition and multiplication by (0.4) and (0.5) requires immediate justification:

one has to prove that these operations are well-defined.

In order to explain that, consider addition first: (0.4) supposedly defines what should be the sum of two equivalence classes $[(m, n)]$ and $[(p, q)]$. However, these equivalence classes are written with the help of a choice of representatives: $[(m, n)]$ is an equivalence class of all pairs (m', n') such that $(m, n) \sim (m', n')$, and (m, n) is only one of them. Pick another one, say, (m', n') such that $(m, n) \sim (m', n')$ but $(m, n) \neq (m', n')$. What happens with our sum defined by (0.4)? It should stay the same simply because

$$[(m, n)] = [(m', n')], \text{ and } [(m', n')] + [(p, q)] = [(m, n)] + [(p, q)].$$

This can be checked directly. Of course, the same applies to the second summand (p, q) .

To sum up, the operations on equivalence classes in (0.4) and (0.5) are defined in terms of arbitrary choices of representatives in these classes. We have to check that the equivalence classes which we produce in the end do not depend on these arbitrary choices. That is what is called well-defined.

Proposition 3. *The operations on $\mathbb{Z} = \mathbb{N} \times \mathbb{N} / \sim$ described in (0.4) and (0.5) are well-defined.*

Proof: Here, we only give a proof for the addition introduced by (0.4). Let $(m, n) \sim (m', n')$ and $(p, q) \sim (p', q')$. That translates to

$$m + n' = m' + n \quad \text{and} \quad p + q' = p' + q \quad (0.6)$$

Then, according to (0.4) we have that

$$[(m', n')] + [(p', q')] = [(m' + p', n' + q')]$$

We have to prove that the result of addition stays unaltered: $[(m' + p', n' + q')] = [(m + p, n + q)]$, or, equivalently, that

$$m' + p' + n + q = n' + q' + m + p,$$

and this immediately follows from (0.6). □

Exercise 1. *Prove that multiplication is well-defined by (0.5).*

Extension of the Operations from $\mathbb{N}$: We have already identified $\mathbb{N}$ as a subset of $\mathbb{Z}$. It would be pretty bad if our newly-introduced operations have nothing to do with the operations of addition and multiplication defined for naturals. Specifically, let $a, b \in \mathbb{N}$. We have identified a with the integer $[(a, 0)]$, and b with $[(b, 0)]$. The sum $a + b$ is again a natural number which we may think about as an integer $[(a + b, 0)]$. It is natural to ask:

do we produce the same integer if we add integers $[(a, 0)] + [(b, 0)]$?

The answer is yes, and it follows immediately from (0.4). This tells that the operation of addition defined by (0.4) extends the addition defined on $\mathbb{N}$. Equivalently, we can say that the operation of addition on $\mathbb{Z}$ defined by (0.4), when restricted to the subset $\mathbb{N} \subset \mathbb{Z}$, coincides with the usual operation of addition of natural numbers.

Proposition 4. *The operations of addition and multiplication on $\mathbb{Z}$ defined by (0.4) and (0.5), when restricted to the subset $\mathbb{N} \subset \mathbb{Z}$, coincide with the usual operations of addition and multiplication on $\mathbb{N}$.*

Proof: Exercise.

Exercise 2. *Prove that the commutative and distributive properties still holds for the addition and multiplication on $\mathbb{Z}$ defined by (0.4) and (0.5).*

1.2 Subtraction on $\mathbb{Z}$:

The goal of the construction of integers was to supply equations (0.1) with solutions. Let us now check that this goal is achieved. Indeed, for any integers

$$a = [(m, n)] \in \mathbb{Z} \text{ and } b = [(p, q)] \in \mathbb{Z}$$

the equation (0.1) has a solution

$$x = [(m + q, n + p)] \in \mathbb{Z}.$$

That is simply because by (0.4) we have

$$x + b = [(m + q, n + p)] + [(p, q)] = [(m + q + p, n + p + q)] = [(m, n)],$$

where the last equality is implied by $(m + q + p, n + p + q) \sim (m, n)$. That allows us to introduce subtraction of integers

$$[(m, n)] - [(p, q)] := [(m + q, n + p)]$$

for any naturals $m, n, p, q \in \mathbb{N}$ as the inverse operation to addition, and write

$$x = a - b$$

for the solution of (0.1).

Remark 3. *One can make use of the cancellation property for naturals*

$$u + v = w + v \Rightarrow u = w$$

to prove that equation (0.1) has a unique solution.

1.3 Traditional Notations:

Of course, in order to simply use integers in the calculations, one does not need to understand the concept of equivalence classes. Instead, one writes negative integers with the minus-sign as $-n$ with a natural n (e.g., -2 , etc.). As we have identified above, natural numbers n with integers $[(n, 0)]$, we now

$$\text{identify negative numbers } -m \text{ with integers } [(0, m)]$$

where $m \in \mathbb{N}$. Using this identification, we will use $\mathbb{Z}$ to denote the set of all integers

$$\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}.$$

Moreover, we can derive from our definition that the product of two negative numbers is positive. For example,

$$(-2)(-3) = [(0, 2)] \cdot [(0, 3)] = [(0 \cdot 0 + 2 \cdot 3, 2 \cdot 0 + 0 \cdot 3)] = [(6, 0)] = 6.$$

and take it for an explanation, not merely a rule.

1.4 Order Relation:

Definition 2. Let S be a set. An **order** on S denoted by “ $<$ ” is a relation on S such that

O1). (Trichotomy) Given any $a, b \in S$, one and only one of

$$a < b, \quad a = b, \quad b < a$$

is true;

O2). (Transition) If $a < b$ and $b < c$, then $a < c$;

We say that S is an **ordered set** if there is an order “ $<$ ” on S .

$\mathbb{N}$ come with an order relation $\leq$: for $a, b \in \mathbb{N}$, we say $a \leq b$ if and only if there exists $c \in \mathbb{N}$ such that $a + c = b$.

We can use it to introduce an order relation on $\mathbb{Z}$:

$$[(m, n)] \leq [(p, q)] \quad \text{if and only if} \quad m + q \leq n + p$$

Exercise:

- 1) Prove that the relation $\leq$ is well-defined.
- 2) Prove that the relation $\leq$ extends the relation $\leq$ on the subset $\mathbb{N} \subset \mathbb{Z}$.
- 3) Prove that the relation $\leq$ on $\mathbb{Z}$ is an order relation.

Summary: Up to now, we have given the construction of integer set $\mathbb{Z}$ with its arithmetic and ordered structure

$$(\mathbb{Z}, +, \cdot, \leq).$$

2. The set of rational numbers $\mathbb{Q}$.

The set $\mathbb{Z}$ also becomes inadequate when division is introduced. A more practical question is:

for $a, b \in \mathbb{Z}$, can we find a solution to the equation $a \cdot x + b = 0$.

Clearly, in general the answer is no. Therefore, we have to find a proper way to enlarge the world of numbers to include all the solutions to general linear equations. Here, we follow a similar idea as we introduce integers to construct the rational numbers.

We consider a set of order pair of integers

$$Q = \{(x, y) \mid x, y \in \mathbb{Z} \text{ and } y \neq 0\}.$$

The first element x of a pair $(x, y) \in Q$ is called the **numerator**, and the second element y is called the **denominator**. Elements of Q are called **numerator-denominator pairs**.

It might be tempting to write (x, y) symbolically as $\frac{x}{y}$. We do not do so at this point because we will define $\frac{x}{y}$ later when considering division, so using $\frac{x}{y}$ now could create confusion. Also, using the notation $\frac{x}{y}$ might lead one to unintentionally use properties about fractions that have not yet been formally proved.

Next, we introduce an equivalent relation in Q :

Definition 3. We say that two elements (x, y) and (z, w) of Q are **rationally equivalent** if $x \cdot w = y \cdot z$, i.e.,

$$(x, y) \sim (z, w) \iff x \cdot w = y \cdot z$$

where $\cdot$ is the multiplication defined on $\mathbb{Z}$.

Proposition 5. The relation $\sim$ is an equivalence relation on the set Q .

Proof. Exercise. The reflexive and symmetry laws involve using the commutative law for $\mathbb{Z}$. The transitivity law requires multiplying an equation by a constant, performing basic manipulations, and using the cancellation law (for a non-zero element of $\mathbb{Z}$). □

Definition 4. If $(x, y) \in Q$ then $[(x, y)]$ denotes the equivalence class containing (x, y) under the above equivalence relation. We define

$$\mathbb{Q} = Q / \sim.$$

In other words,

$$\mathbb{Q} = \left\{ [(x, y)] \mid x, y \in \mathbb{Z} \text{ with non-zero } y \right\} / \sim.$$

2.1 Addition and Multiplication.

We define the usual operations on $\mathbb{Q}$ as follows: let $[(x, y)]$ and $[(z, w)]$ be elements of $\mathbb{Q}$. Then addition $+$

$$[(x, y)] + [(z, w)] = [(xw + yz, yw)] \tag{0.7}$$

multiplication $\cdot$

$$[(x, y)] \cdot [(z, w)] = [(xz, yw)] \tag{0.8}$$

Similar to the case for the integers, these definitions involve equivalence classes, so we must show that they are well-defined. We leave those justifications as an exercise. For example, to verify that *addition on $\mathbb{Q}$ is well-defined*:

Proof. We want to show that if $[(x, y)] = [(x', y')]$ and $[(z, w)] = [(z', w')]$ then

$$[(xw + yz, yw)] = [(x'w' + y'z', y'w')].$$

In other words, suppose $(x, y) \sim (x', y')$ and $(z, w) \sim (z', w')$. We must show $(xw + yz, yw) \sim (x'w' + y'z', y'w')$. By definition of $\sim$, we need to show that

$$(xw + yz)(y'w') = (x'w' + y'z')(yw).$$

Similarly, we can state the following propositions without proof:

Proposition 6.

- i). *Multiplication on $\mathbb{Q}$ is well-defined.*
- ii). *Addition in $\mathbb{Q}$ is commutative.*
- iii). *Multiplication in $\mathbb{Q}$ is commutative.*

Proposition 7. *For any $[(x, y)] \in \mathbb{Q}$,*

$$[(x, y)] + [(0, 1)] = [(x, y)] \quad \text{and} \quad [(x, y)] \cdot [(1, 1)] = [(x, y)].$$

This proposition implies

- $[(0, 1)] \in \mathbb{Q}$ is the identity for addition;
- $[(1, 1)] \in \mathbb{Q}$ is the identity for multiplication.

2.2 Division.

Using definition (0.8), we can easily compute that

$$[(x, y)] \cdot [(y, x)] = [(1, 1)] \quad \text{for any non-zero integers } x, y.$$

This implies that

for any non-zero $x, y \in \mathbb{Z}$, the multiplication inverse of $[(x, y)]$ is $[(y, x)]$,

and we denote the multiplication inverse $[(y, x)]$ as $[(x, y)]^{-1}$.

The existence of the multiplication inverse leads us to introduce the **division** operation: for nonzero $z, w \in \mathbb{Z}$, we define

$$[(x, y)] / [(z, w)] := [(x, y)] \cdot [(z, w)]^{-1}. \quad (0.9)$$

2.3 Identifying $\mathbb{Z}$ as a subset of $\mathbb{Q}$.

The identification can be done through the canonical embedding $\mathbb{Z} \rightarrow \mathbb{Q}$ defined by the rule

$$a \mapsto [(a, 1)].$$

If we identify $a \in \mathbb{Z}$ with its image in $\mathbb{Q}$, then we can think of $\mathbb{Z}$ as a subset of $\mathbb{Q}$. So from now on, if $a \in \mathbb{Z}$, we will think of a and $[(a, 1)]$ as being the same element of $\mathbb{Q}$. By this convention, $0 \in \mathbb{Z}$ is identified with $[(0, 1)]$, and $1 \in \mathbb{Z}$ is identified with its image $[(1, 1)]$. By a propositions above,

$0 \approx [(0, 1)]$ is the additive identity and $1 \approx [(1, 1)]$ is the multiplicative identity as expected.

Since we now think of $\mathbb{Z}$ as a subset of $\mathbb{Q}$, we have to be careful with $+$ and $\cdot$ in $\mathbb{Z}$. We defined these operations for $\mathbb{Z}$ in one way previously, and then defined them for $\mathbb{Q}$ in a different way. Do we get the same answer for integers $a, b \in \mathbb{Z}$ as for the corresponding elements $[(a, 1)]$ and $[(b, 1)]$ in $\mathbb{Q}$? The answer is yes since

$$a + b \approx [(a, 1)] + [(b, 1)] = [(a + b, 1)] \approx a + b$$